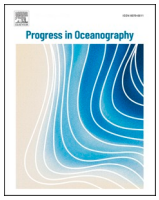
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Challenging one-way fish modelling: two-way coupling reveals critical marine ecosystem dynamics

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ABSTRACT

Anthropogenic stressors such as climate change and fishing are altering marine ecosystem dynamics. To understand and predict how marine foodwebs are changing, both top-down and bottom-up trophic dynamics need to be included in our model systems. A common simplification employed is a one-way coupled modelling approach, where results of lower trophic level (LTL) (plankton) models are used to drive fish models, but feedbacks due to fish predation are not applied to the LTL model. Thus, fish do not perceive the resulting reduction in prey biomass, limiting the predictive capacity of the models. Here we have implemented innovative two-way coupling between an LTL model, ERSEM, and the community size spectrum fish model configuration of MIZER, to produce a complete model chain from photons to fisheries. The model is used to investigate the impact of two-way coupling on the North West European Shelf ecosystem. Within the two-way coupled model, clearance rates of fish and mortality rates of mesozooplankton had to be reduced, highlighting limitations of parameterisations within one-way coupled models. Mesozooplankton was the most sensitive group to two-way coupling with a 17% decrease in biomass across the shelf in addition to changes in phenology. Furthermore, two-way coupling decreased carbon cycling across the shelf driven by the top-down pressures on mesozooplankton reducing their production of organic matter and weakening the microbial loop. This impacted the ocean's ability to absorb atmospheric carbon and consequently has climate change implications. Two-way coupled fisheries models should therefore be considered in the next generation of earth system models.

1. Introduction

Fish provide a critical food supply to the human population. However, fishing and climate change are modifying fish populations across the world's oceans (Allison et al., 2009; Bindoff et al., 2019). Climate change may modify nutrient concentrations (Holt et al., 2016; Kwiatkowski et al., 2020; Wakelin et al., 2015) with impacts cascading up through the food web to higher trophic levels (Cheung et al., 2011; Lotze et al., 2019). Likewise, fishing and increasing temperatures alter fish biomass and migratory patterns, modifying top-down pressures on other fish and plankton biomasses (Lynam et al., 2017; Thorpe et al., 2022). The balance between these top-down and bottom-up processes controls the biomass of plankton and fish in the ocean (Lynam et al., 2017). To make predictions on future fish stocks as well as the ocean's ability to absorb carbon dioxide (CO_2), it is imperative to understand the ecosystem interactions between these bottom-up and top-down processes controlling both plankton and fish biomasses. However, these processes are not well represented in many models.

End-to-end models that cover physics to fish are required to disentangle these complex foodweb processes. In particular, end-to-end models with two-way feedbacks between the lower trophic levels (LTL) and higher trophic levels (HTL) are required to properly understand the bottom-up and top-down drivers in the marine ecosystems and to make accurate predictions on the state of the marine ecosystem (Rose, 2012) and carbon cycling. Previously, a one-way coupled approach has been used in a number of modelling frameworks to ease model implementation and reduce computational costs. When a one-way coupled approach is used, a HTL model is forced by outputs of an existing LTL model, such as plankton biomass, net primary production and chlorophyll concentrations (Blanchard et al., 2009; Lotze et al., 2019; Wilson et al., 2021). As both models are run separately, there are no feedbacks from fish predation on the plankton biomass within the LTL, nor on the

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dissolved or particulate constituents through excretion and mortality. Thus, top-down trophic cascades within the ecosystem cannot be reproduced. This increases the uncertainty of the assessment of the impact of fishing or climate change within those models.

The lack of explicit predation by fish on plankton also has implications for LTL models. For example, no publicly available reanalyses for the Copernicus Marine Monitoring Service (https://marine.copernicus.eu/) include coupling with HTLs and thus are missing the dynamical response of plankton to fish within the model. In addition, all global climate models contributing to the latest Coupled Model Intercomparison Project (CMIP6) do not include feedback of fish on plankton (e.g. Séférian et al., 2020), limiting predictions of the ocean's capability of absorbing atmospheric CO_2 (Cavan and Hill, 2022). Typically, LTL models have a top closure term, such as intraguild predation by mesozooplankton, or an additional mortality term to represent the feeding of HTLs, but these are generally poorly tested and may produce different seasonal dynamics to an explicit representation of fish predation. The simplistic top closure term limits a model's ability to predict future carbon budgets and carbon sequestration by the marine system.

There is increased recognition of the need for improved end-to-end models with suitable ecosystem complexity, and spatial and temporal scales (Horn et al., 2021). The lack of two-way coupling in FishMIP models was hypothesised to lead to greater divergence in model ensemble results, than if two-way coupled models were used, due to contrasting trends of production and biomass variables fed as input from LTLs (Heneghan et al., 2021). Similarly, the divergence in fish biomass between regional and global models was in part was hypothesised to be due to more complex interactions in some regional models (Eddy et al. 2025), indicating the need for improved end-to-end models both regionally and globally. In addition to impacts on the carbon cycle, fishing is increasingly shown to impact local to global biogeochemical cycles through the removal of phosphorus and nitrogen (González Ortiz et al., 2025; Ray et al., 2025) but these impacts cannot be investigated with models that are one-way coupled.

The issues listed above raise key questions about the uncertainty within existing ecosystem models for understanding how anthropogenic pressures and climate change will alter fish communities and the carbon cycle. Trophic interactions are complex and often hard to predict (Heath et al., 2014). However, at present, climate change impacts on fish populations are largely modelled using zooplankton prey fields that are entirely divorced from the fish populations which feed on them with feedbacks also poorly represented in the LTL model. There is therefore a pressing need to understand if a more realistic representation of interactions between fish and zooplankton will result in models that provide different conclusions about the impacts of climate change and anthropogenic pressures on fisheries.

A two-way coupled end-to-end modelling approach is however becoming an increasingly popular tool to model marine ecosystems, although global applications of two-way coupling are very limited compared to one-way coupled models. Initial two-way coupled frameworks, such as Atlantis (Audzijonyte et al., 2019; Fulton et al., 2011; Fulton et al., 2005; Ortega-Cisneros et al., 2018), ROMS-$N_2P_2Z_2D_2$-OSMOSE (Travers et al., 2009; Travers-Trolet et al., 2014), EwE (Christensen and Walters, 2004), and StrathE2E (Heath et al. 2021) amongst others (Fennel, 2008, 2009; Megrey et al., 2007) typically lacked high spatial and vertical resolutions or included a relatively simple representation of lower trophic level dynamics. Nonetheless, these models highlighted the importance of two-way coupling, showing the cascading effects of bottom-up and top-down inputs (Travers et al., 2009), the importance of alternative trophic pathways and diet matrices between models (Shin et al., 2010; Smith et al., 2015) and impacts of anthropogenic activities such as fishing on the entire ecosystem and not just target species (Smith et al., 2011). Such models can be used to suggest management strategies that consider trade-offs throughout the entire system (Forrest et al., 2015) such as minimising impact of management scenarios on yield whilst achieving a high level of ecosystem improvement (Smith et al., 2011) and having management strategies that target the entire ecosystem rather than only one particular species of fish (Fulton and Smith, 2004).

More recently, models have been coupled to state-of-the-art hydrodynamic models, and applied to small local domains (Bănaru et al., 2019; Diaz et al., 2019; Gkanasos et al., 2021; Rose et al., 2015; Travers et al., 2009; Travers-Trolet et al., 2014) with only a handful applied to regional (Baltic and North Sea; Daewel et al., 2019; Bering Sea; Ortiz et al., 2016; Baltic; Radtke et al., 2013) and global domains (Aumont et al., 2018). These models encompass a range of complexity across both the higher and lower trophic level, e.g. they vary in the number of plankton and fish species they represent. However, only a limited number of these complex 3D models have looked at the impact of two-way vs one-way coupling on the entire carbon cycle including fluxes from fish to plankton to nutrients.

To explore the impact of two-way feedbacks throughout the marine ecosystem, we couple the community size-spectrum version of the HTL model MIZER (Blanchard et al., 2009, 2012; Scott et al., 2014) to the LTL model ERSEM (Butenschön et al., 2016). Both the community and species-specific versions of MIZER have been extensively used to investigate fish biomass and consequences of environmental and anthropogenic changes within the marine system (Blanchard et al., 2009, 2012) but only in a one-way coupled setting where there is no impact of fish on the LTL model. The innovative two-way coupling approach introduced here means that fish predation will impact plankton biomass in addition to producing organic detritus, inorganic nutrients and carbon dioxide, through respiration and excretion.

The model is implemented on the North West European Shelf using a 3D hydrodynamical model and run from 1981 to 2017. Shelf seas are important regions of the global ocean, supporting high rates of primary productivity through river inputs and exchanges with high nutrient off shelf water. The high rates of primary production supports fish production with approximately 90% of fish catches occurring in shelf and coastal waters globally (Pauly et al., 2002). The North West European Shelf is extensively fished (Kovšars et al., 2022) with large consequences to the marine food web. In addition, it is important in the global carbon cycle through its role in acting as a sink of atmospheric CO_2 (Kitidis et al., 2019; Laruelle et al., 2014, 2018; Legge et al., 2020). Historically, specific areas of the North West European Shelf such as the North Sea, have undergone regime shifts within the marine food (e.g. Beaugrand, 2004; Blöcker et al., 2023; Djeghri et al., 2023; Sguotti et al., 2022) indicating the shelf's sensitivity to environmental and anthropogenic pressures.

To investigate the impact of two-way coupling, we will run three simulations: the first two run ERSEM and MIZER separately whilst the final simulation will include two-way coupling. In this study, we will show the predicted spatial distribution of fish biomass and landings across the North West European shelf in addition to showing the impacts of two-way coupling on the LTL and HTL ecosystems, including plankton and fish biomasses and air-sea and seafloor carbon fluxes. Finally, we will discuss the implications of the differences in parameterisation required to successfully model the entire ecosystem.

2. Methods

2.1. Modelling system description

In order to investigate the impact of two-way coupling between a LTL and HTL model, a fully coupled 3D modelling system was used that includes the hydrodynamical model NEMO (Madec and the NEMO team, 2016), the LTL biogeochemical model ERSEM (Butenschön et al., 2016) and the HTL fish Community Size Spectrum Model configuration of MIZER (Blanchard et al., 2009, 2012, 2014; Scott et al., 2014). The models are coupled together using the Framework for Aquatic Biogeochemistry Model (FABM; Bruggeman and Bolding, 2014). The model was run using the UK Archer2 National Computing service (Beckett

et al., 2024).

2.1.1. Model domain and hydrodynamic model

The hydrodynamic model, NEMO is a general circulation model. Here we run version 3.6 with the setup described in O'Dea et al. (2017). The models were run on the Atlantic Margin Model (AMM7) domain which covers the North West European Shelf and part of the East Atlantic Ocean with the domain spanning from 40°N to 65°N and from 20°W to 12°E at a spatial resolution of approximately 7 × 7 km (Fig. S1). A z*-s coordinate system is used with 51 levels which are thinner on the shelf where depth is shallower and thicker in the open ocean.

2.1.2. Biogeochemical (lower trophic level) model

ERSEM is a biogeochemical model which uses a functional group approach to represent plankton and benthic biomass. ERSEM simulates the cycling of five elements: carbon, nitrogen, phosphorus, silicon and oxygen and specifically includes four phytoplankton groups, and three zooplankton groups, which represent different functional types and size classes (Table 1). ERSEM also includes bacteria, three different classes of dissolved organic matter (DOM) characterised by different lability and three different size classes of detrital particulate organic matter (POM). In addition to a pelagic ecosystem, ERSEM also comprises of a benthic component which simulates the cycling of elements in the aerobic and anaerobic benthic system through nutrients, POM, DOM, meiofauna, deposit- and suspension feeders and bacteria. In this version of the model, total alkalinity is assigned as a prognostic state variable rather than semi-diagnostic as in Butenschön et al. (2016).

2.1.3. Fish (higher trophic level) model

The community size spectrum version of MIZER (Blanchard et al., 2009, 2012) represents fish at a community-level. Size is often considered to be a 'master' trait, controlling the rates of physiological processes such as growth and respiration (Andersen et al., 2016; Brown et al., 2004). Many ecological processes also scale with size, in particular predation (Dunic and Baum, 2017). By explicitly representing size, size spectrum models provide a simple but effective tool to understand ecosystems (Heneghan et al., 2019). Here, MIZER has been configured to be equivalent to the pelagic component of the Community Size Spectrum Model (Blanchard et al., 2009, 2012) and does not include feeding on the benthos. The primary reason for this is that ERSEM's detailed benthic submodel lacks any representation of organism size in its description of its dominant types of benthic fauna, deposit feeders and suspension feeders. Thus, it is non-trivial to determine what fraction of the biomass of these two benthic groups might be predated upon by different size classes of fish. The lack of benthic feeding does imply that demersal fish species are not represented; this limitation will be revisited in the discussion section.

2.2. FABM-MIZER implementation

As part of this study MIZER has been recoded into FABM (Bruggeman, 2021), resulting in a fully online coupled system with two-way feedbacks between the HTL and LTL models. Additionally, the FABM framework allows FABM-MIZER to easily be coupled to a wide variety of hydrodynamical and biogeochemical models. Here we provide a brief overview of the FABM implementation of MIZER and the coupling to ERSEM. A full description of the implementation can be found in the Supplementary Material with Fig. 1 summarising the coupling process.

FABM-MIZER was run as a 2D depth-integrated model in the same spatial domain and horizontal grid as the hydrodynamic-biogeochemical model. The coupling between the 3D NEMO-FABM-ERSEM and 2D FABM-MIZER was carried out by distributing simulated fish biomass throughout the water column at every time step (in our case every 5 min), under the assumption that the vertical distribution of fish mirrors that of the plankton prey in the LTL. This keeps computational costs low, and avoids making assumptions about how a community of fish will swim in the vertical. Similar simplifications have been made with other two-way coupled models run with regional 3D hydrodynamical models such as ECOSMO-E2E (Daewel et al., 2019), ERGOM + fish (Radtke et al., 2013) and BESTNPZ-FEAST (Ortiz et al., 2016). Through the assumed vertical distribution of fish, we infer depth-resolved rates of predation on plankton, waste production and respiration for the fish community. These rates are subsequently applied to the corresponding depth-resolved ERSEM variables to establish the two-way coupling (Fig. 1). In order to bring the model into FABM and enable easy coupling with other LTL models, the units for fish biomass are in mmol carbon m^{-2}. To convert to grams of wet mass (WM), which are the original units used by Blanchard et al., (2009), and are often used in observational data and other models, we used conversion factor of 0.12 gWM:mmol carbon, assuming 10% of fish WM is carbon (Boudreau and Dickie, 1992) and using the molar mass of carbon.

The community size-spectrum model was run with 100 size classes of fish ranging from 10^{-3} to 10^{6} g. Model equations describe the growth of each size class of fish and various loss terms associated with predation and natural mortality. To incorporate the community size-based fish model with FABM, the source terms for each size class of fish, (as well as prey and waste) are computed, and provided as-is to the hydrodynamic model to be time-integrated using the same solver it uses for all other model variables.

Within this implementation, prey capture and ingestion are formulated as a linear functional response without saturation or maximum (cf. Blanchard et al., 2014). Metabolic losses (e.g., maintenance) are implicitly accounted for by using a low efficiency (0.2) for prey-to-biomass conversion, rather than by an explicit standard metabolism loss (cf. Blanchard et al., 2014). In order to couple to ERSEM, nitrogen, phosphorus and silicon have to be included in the equations in addition to carbon. The fish uptake these nutrients from their prey of which a fraction is assimilated. Somatic growth is limited by the minimum of the assimilated carbon, nitrogen or phosphorus converted to carbon using the Redfield ratio (Redfield et al., 1963). These ratios are based on conventions in marine biogeochemical modelling. While they fall within the range of ratios observed by Sterner and George (2000), they could be modified in the future to account for the tendency of higher trophic levels of be richer in phosphorus (Le Mézo et al., 2022; Sterner and Elser, 2002). Any excess assimilated carbon or nutrients, as a result of keeping fish biomass at a Redfield ratio, are egested and contribute to the detritus pools tracked by the LTL, in this case the large size class of detritus within ERSEM. Silicon is not incorporated into fish biomass and is instantaneously egested.

As we are describing a community of fish, rather than a single species, there is no straightforward relationship between egg production and the biomass in larger ("adult") size classes – reproductive effort per size class would depend on species composition, which is unknown. Instead of explicitly modelling reproduction, egg production is therefore dictated by the presence of plankton prey. Specifically, similar to Blanchard et al., (2009), the density of the smallest size class of fish (1 mg) is relaxed towards an expected density that is obtained by extrapolating the depth-integrated plankton size spectrum, using sizes

Table 1
ERSEM plankton functional types and their assigned mass ranges to produce plankton size spectrum used by FABM-MIZER. (Letters/numbers in brackets are native ERSEM naming convention). ESD = Estimated spherical diameter.

ERSEM group	Size (wet mass in 10^{-12} g = 1/6 π ESD in μm)
Picophytoplankton (P3)	0.2–2 μm ESD
Nanophytoplankton (P2)	2–20 μm ESD
Microphytoplankton (P4)	20–200 μm ESD
Diatoms (P1)	20–200 μm ESD
Heterotrophic nanoflagellates (Z6)	2–20 μm ESD
Microzooplankton (Z5)	20–200 μm ESD
Mesozooplankton (Z4)	200 μm ESD–1 mg

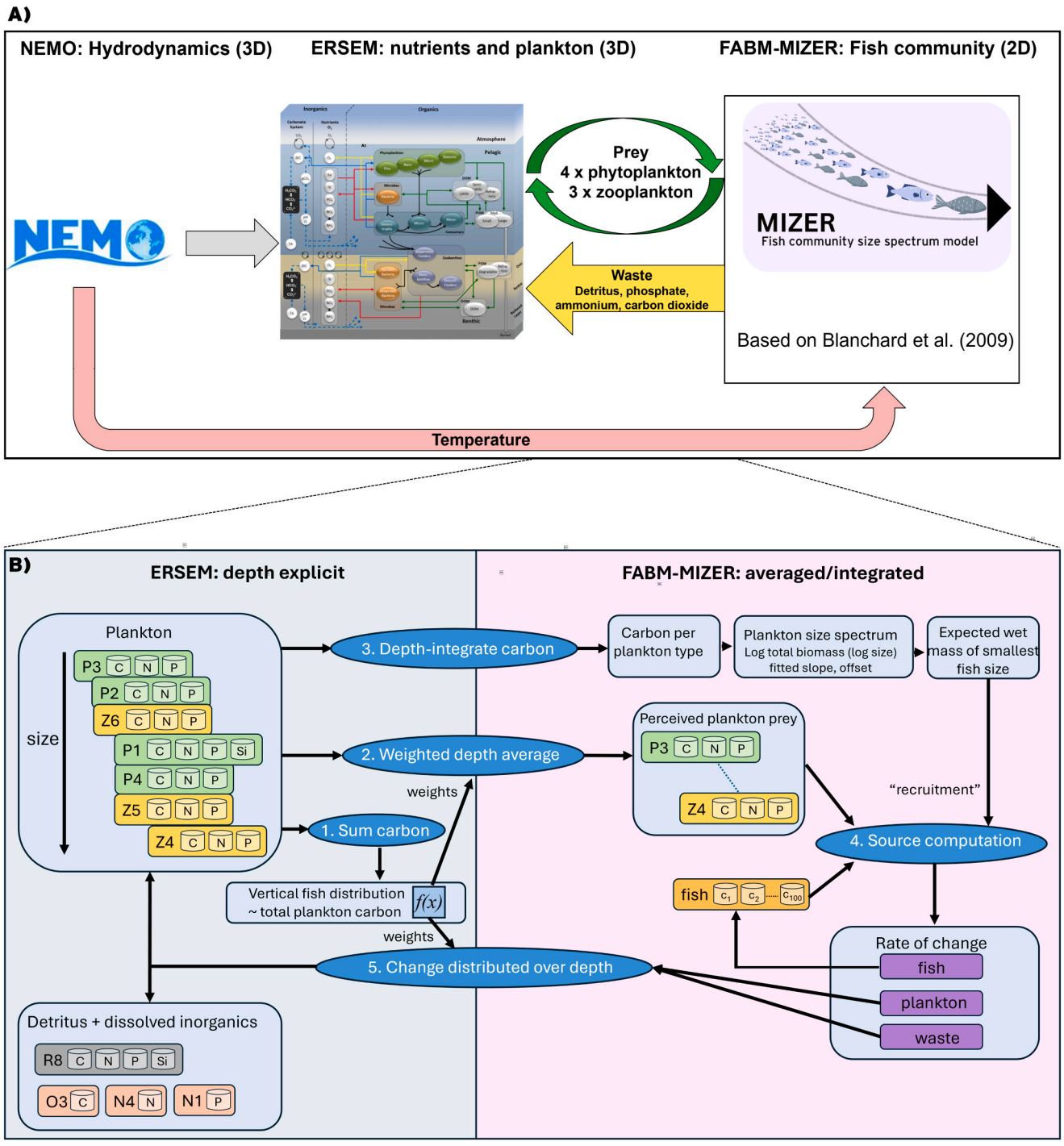


Fig. 1. (A): Model schematic showing the innovative two-way coupling between NEMO, ERSEM and FABM-MIZER implemented for the first time in this study. (B) Detailed representation of coupling process and methods between ERSEM and FABM-MIZER. State variables are indicated by cylinder shapes, processes by oval shapes. Green boxes represent phytoplankton (P), yellow zooplankton (Z), names follow native ERSEM naming convention (see Table 1). A full description of the coupling process can be found in the supplementary material.

assigned in Table 1. The scale factor and exponent of this spectrum are computed at run time by linear regression. If the expected density of the smallest size class is greater than the current density, the latter is relaxed towards the expected value at a rate of 5 d^{-1}. If the expected density is smaller, relaxation is not applied. In this manner, we ensure that effective "recruitment" is always positive. In order to achieve mass balance, any excess/supply of carbon/nitrogen/phosphorus needed for recruitment is returned/taken from the detrital POM pool, in this case large POM.

To include the impact of fishing, we follow Blanchard et al. (2012) and Scott et al. (2014) in prescribing knife-edge fishing where a constant rate of fishing occurs (F, yr^{-1}) for all fish over a certain size (w_f) on the entire domain. Fishing pressure can vary between 0.2 and 1 yr^{-1} depending on the sustainability of fishing (Blanchard et al., 2012; Scott et al., 2014). Likewise, the minimum fish size (w_f) landed varies between 1.25 g (Blanchard et al 2012) to 1000 g (Scott et al., 2014). Here we follow Blanchard et al. (2012) in assigning a fishing pressure of 0.8 yr^{-1} but use a minimum size of fish which is fished of 80 g WM, based on five

year test simulations using a range of F and w_f values, and comparing predicted fish landing mass against observations (Table S3). Assigning a w_f of 80 g represents all fish caught for human consumption, but potentially misses the fish caught for fishmeal, which may be important in the North Sea. Fish landings reported here do not take into account any legal quotas on fishing and is purely a function of fish wet mass present in the model. Biomass removed through application of the fisheries mortality is tracked as "landings" that no longer interact with the ecosystem. This reduces the quantity of bioavailable carbon, nitrogen and phosphorus within the system.

2.3. Simulations

In total three main simulations were undertaken from 1981 to 2017 for the North West European Shelf, to assess the impact of two-way coupling on both fish biomasses and LTL carbon cycle compared to one-way coupling. These are: 1) *Baseline LTL simulation*: simulation where standard ERSEM was run with NEMO-FABM 2) *offline HTL simulation*: Plankton and temperature from the baseline LTL simulation were used as forcing conditions for the FABM-MIZER simulation which was run separately and thus with only one-way coupling. For each cell in the model domain, a spin-up of 50 years was undertaken for MIZER using average plankton biomass over 1981–2017 before running using 1981–2017 forcings. The one-way coupled simulation also quickly provides reasonable initial fish biomass to run the two-way coupled mode. 3) *Two-way coupled simulation*: NEMO-FABM-ERSEM-MIZER with two-way feedbacks between plankton and fish, and production of detritus and nutrients/CO_2 by fish. Biomass for each size class of fish from the last day of the offline HTL simulation was used as initial conditions (January 1981) for the two-way coupled AMM7 simulation. All other boundary conditions and initial conditions remained exactly the same as the baseline LTL simulation with the exception of the removal of some mesozooplankton mortality parameters (natural + intra-guild predation) as described in section 2.4.1.1. Results are presented for 1986–2017 to allow a further 5 years of model spin-up.

2.4. Model setup

2.4.1. Parameterisation

2.4.1.1. FABM-MIZER parameters. Here, the Community Size Spectrum Model version of MIZER was implemented and thus parameterisations generally follow Blanchard et al. (2009) with key parameters and exceptions described in the main text, Supplementary Material and Table S1.

In order to achieve satisfactory results upon two-way coupling and enable mesozooplankton and consequently fish biomass to not be substantially reduced, the parameters associated with volumetric search rate of food (Equation S8) were modified from Blanchard et al. (2009) using data from Acuña et al. (2011) (Fig. S2) which alleviates the predation pressure on mesozooplankton. The values used deviate from those directly calculated from the clearance rate equations for fish reported by Acuna et al., as we included both filter feeding and searching fish in the analysis whilst limiting the fish sizes to those matching our definition within MIZER (10^{-3}–10^{6}g WM). Various other permutations of the volumetric search rate were investigated based on literature reported values (Blanchard et al., 2012; Kiørboe and Hirst, 2014) as shown in Fig. S2 and Table S2. Parameters derived from data in Acuña et al. (2011) (Fig. S2) were the only set which did not detrimentally impact mesozooplankton and consequently fish biomass (Fig. 2).

2.4.1.2. ERSEM parameters. The baseline LTL simulation uses parameters from Butenschön et al. (2016). The two-way coupled simulation uses the same ERSEM parameterisation except for the handling of mesozooplankton mortality. Mesozooplankton intra-guild predation (often termed cannibalism) was replaced with explicit fish predation. In addition, to achieve satisfactory mesozooplankton biomass, mesozooplankton basal and over wintering mortality, which are linear functions of mesozooplankton biomass, were turned off in the two-way coupled simulation, although mortality due to lack of oxygen remained.

2.4.2. Boundary and initial conditions

Physical and biogeochemical boundary and initial conditions were

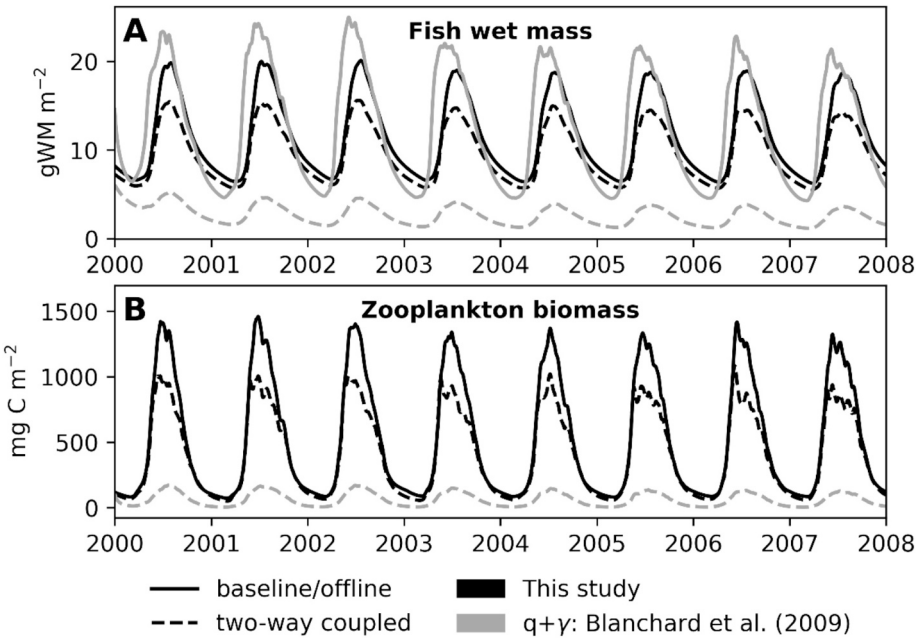


Fig. 2. Comparison between the spatial average of (A) fish wet mass and (B) zooplankton carbon biomass across the North West European Shelf as a result of changing clearance parameters (q and gamma; equation S8) used in this study (black lines) versus gamma and q from Blanchard et al., (2009) (grey lines) in the offline HTL/baseline LTL simulation (solid lines) and two-way coupled simulation (dashed lines). Note fishing parameters were different between the two simulations but this does not impact overall trend.

derived from the best available datasets representing the North West European Shelf marine ecosystem. The baseline LTL and two-way coupled simulations are forced by hourly ERA5 atmospheric data (Hersbach et al., 2020). Physical ocean boundary conditions are taken from the GLORYS reanalysis (Ferry et al., 2012) while biogeochemical boundary conditions are taken from the World Ocean Atlas for nutrients and oxygen (Boyer et al., 2018) and GLODAP dataset for carbonate chemistry (dissolved organic carbon (DIC) and total alkalinity; Lauvset et al. (2016)). Compared to previous implementations of ERSEM (Artioli et al., 2012, 2014), total alkalinity has been considered a fully prognostic variable affected by changes in nutrient concentration due to biogeochemical processes and external inputs. GLODAP climatology is representative of 2002 values (Lauvset et al., 2016). A linear annual trend calculated from the insitu GLODAP dataset was applied to the climatological DIC and total alkalinity data to represent the changes in atmospheric CO_2 with time. In addition, a seasonal cycle was superimposed on total alkalinity and DIC to include the effect of primary production on the carbonate chemistry. The seasonal cycle of nitrate from the World Ocean Atlas (Boyer et al., 2018) was used and transformed to units of total alkalinity and DIC (using Redfield ratio). Low phytoplankton biomasses and detrital particulate organic carbon (POC) concentrations were assigned at the Atlantic boundary to avoid spurious exponential growth of phytoplankton or sinking organic matter, occurring at the boundary due to numerical instabilities. In addition, semi-labile and semi-refractory dissolved organic carbon (DOC) concentrations were defined at the boundary following Powley et al., (2024).

Time varying daily river discharge, nutrient loads (nitrate, ammonia, phosphate and silicate), total alkalinity and oxygen are produced from an updated version of the riverine dataset used in Lenhart et al. (2010) combined with climatology of daily discharge data from the Global River Discharge Data Base (Vorosmarty et al., 2000) and from data prepared by the Centre for Ecology and Hydrology as used by Young and Holt (2007). Atmospheric deposition of nitrate and ammonia are applied to the surface ocean using data from the Environmental Monitoring and Evaluation (EMEP) model (Simpson et al., 2012).

The initial conditions for the baseline LTL simulation were the same as in Powley et al., (2024), who investigate the impact of terrestrial dissolved organic carbon across the North West European Shelf. A spin-up period of 5 year repeating forcing between 1981–1985 for two cycles was undertaken due to differences in meteorological forcing and some boundary conditions between this simulation and the one undertaken by Powley et al. The initial conditions for the offline HTL and two-way coupled model are stated in section 2.3.

2.5. Validation of simulated fish landings

Spatial validation for fish in the two-way coupled simulation was undertaken using landing data from STECF/ICES (ICES, 2021a, b, 2023; Kovšars et al., 2022). Annual averaged data was taken for the same years covered by the model simulations where possible. Pelagic fish were extracted from the observational datasets using the same classification for pelagic fish as in Heath et al. (2021), and updated for fish species not present in the Heath dataset using FAO (2024) together with FISHBASE environment classifications (Froese et al., 2024). Overall, over 98% of biomass was classified in the dataset. We have not attempted to validate the size spectrum of fish in this work as this would require detailed information on fish catch which typically comes from trawl surveys (i.e. International Bottom Trawl Survey; https://www.ices.dk/community/groups/Pages/IBTSWG.aspx). Considering this study does not represent demersal fish, it is not prudent to try and compare the predicted size spectrum with this data.

The LTL components and physical components of the model were validated for both simulations based on the gridded North Sea Biogeochemistry Climatology (Hinrichs et al., 2017; nutrients, oxygen, chlorophyll and salinity) and OSTIA (E.U Copernicus Marine Service Information, 2023; temperature) datasets.

3. Results

3.1. Validation of model

3.1.1. Fish biomass

Fish biomass estimates in the North Sea in both simulations (Table 2; Fig. 3) were comparable to observed data and other predictions of fish biomass from models. Pelagic fish biomass was estimated to be 10–11 gWM m^{-2} across the North Sea from fish stock data (Greenstreet et al., 1997; Mackinson &and Daskalov, 2007) compared to 12.0 and 9.5 g WM m^{-2} predicted by the offline and two-way coupled simulations, respectively (Table 2). Similarly, predicted total integrated fish biomass across the North Sea of 0.51 and 0.40 $\times$ 10^6 t C in the offline and two-way coupled simulation are comparable to that predicted by the end-to-end model ECOSMO-E2E of 0.462 $\times$ 10^6 t C (Daewel et al., 2019). However, there is a distinct difference in the spatial distribution of fish between our simulations and ECOSMO-E2E, with Daewel et al., (2019) indicating substantial fish biomass in the Norwegian Trench, and nearby areas relative to the rest of the North Sea. In this study, there is more fish biomass in the Central and Southern North Sea than Norwegian Trench (Table 2; Fig. 3) which may partially be a consequence of the boundary condition applied for the Baltic Sea in this study.

3.1.2. Fish landings

Uncertainties in observed predictions of total fish stocks are high due to the inability to survey all fish in the ocean and capture all species. Fish landings on the other hand are required to be reported and provide another validation metric for fish models. There is still uncertainty in fish landing observations, as data may not be reported for all countries or be sufficiently specific for pelagic fish to be extracted from it. Both the offline and two-way coupled model captured the important role of the Southern North Sea for fish landings with observed fish landings in the Celtic Sea also comparable to model predictions (Fig. 4 and S5). High fish landings near to the shelf edge and near the north west coast of Scotland (Fig. 4 and S5) were not predicted by either simulation. Across the North West European Shelf fish landings were predicted to be 1.5 WM m^{-2} yr^{-1} (1.6 $\times$ 10^6 t yr^{-1}; Table 2) and 1.0 gWM m^{-2} yr^{-1} (1.0 $\times$ 10^6 t yr^{-1}) in the offline and two-way coupled simulations, respectively. For the North Sea, the offline and two-way coupled simulation mean predicted fish landings over 1986–2017 were 1.7 and 1.0 gWM m^{-2} yr^{-1} (0.7 and 0.4 $\times$ 10^6 t yr^{-1}) respectively in comparison to observed landings of pelagic fish of ~1.8 gWM m^{-2} yr^{-1} (1.2 $\times$ 10^6 t yr^{-1}) (ICES, 2021b). However, some degree of underestimation of total North Sea landings by the model is to be expected, as the area from which observational data are reported extends beyond the model domain towards the Baltic. Both simulations predicted fish landings that were substantially lower than the modelled and observed pelagic fish landing data reported for the North Sea in Heath (2012) of 2.1–2.4 and 2.7–3.0 gWM m^{-2} yr^{-1} respectively (calculated from gC to gWM using conversion factors of 0.162–0.184 gC $(gWM)^{-1}$ in Heath (2005)). Although the offline simulation is closer to observations for the North Sea than the two-way coupled simulation this is partially due to a greater overestimation in some cells within the offline simulation than the two-way coupled (Fig. 4) which balances out the cells which are underestimated leading to a seemingly better validation. In the Celtic Sea ICES ecoregion, the offline model predicted the same landings as those observed of 0.78 gWM m^{-2} yr^{-1} (703 $\times$ 10^3 t yr^{-1}) (ICES, 2021a) while the two-way coupled model slightly underestimated landings at 0.60 gWM m^{-2} yr^{-1} (542 $\times$ 10^3 t yr^{-1}).

3.1.3. Lower trophic level model

ERSEM has been previously used in multiple studies before (Lessin et al., 2020; Powley et al., 2024; Wakelin et al., 2020, 2012) and hence a comprehensive validation is not undertaken here. However, two-way coupling does not substantially change the overall model validation over the course of the model simulation for dissolved nutrients,

Table 2
Simulated fish wet mass (bold) and yearly fish landings across key geographic regions (mapped in Fig. S1), reported both normalised to surface area and as total mass, averaged over 1986–2017. Numbers can be converted to carbon units by multiplying by 0.1 (assumes 10% of wet mass is carbon). Note these regions have different boundaries to the ICES ecoregions.

			Normalised to surface area		Total mass		
	Surface Area (10^3 km^2)	Type	Offline HTL (g WM m^{-2})	Two-way coupled (gWM m^{-2})	Offline HTL (10^6 t WM)	Two-way coupled (10^6 t WM)	% change
Southern North Sea	**93.8**	**Biomass**	**13.6**	**9.3**	**1.3**	**0.87**	**−31.7**
		Landings	2.7	1.5	0.25	0.14	−43.3
Central North Sea	**148.3**	**Biomass**	**13.3**	**10.4**	**2.0**	**1.5**	**−21.7**
		Landings	1.9	1.2	0.29	0.17	−39.1
Northern North Sea	**178.9**	**Biomass**	**10.2**	**9.0**	**1.8**	**1.6**	**−12.1**
		Landings	0.95	0.63	0.17	0.11	−33.5
Channel	**72.1**	**Biomass**	**13.5**	**10.9**	**0.97**	**0.79**	**−19.2**
		Landings	2.1	1.4	0.15	0.098	−34.2
Skagerrak	**35.3**	**Biomass**	**12.2**	**8.6**	**0.43**	**0.30**	**−29.1**
		Landings	2.1	1.1	0.076	0.039	−48.9
Norwegian Trench	**54.6**	**Biomass**	**11.6**	**9.5**	**0.63**	**0.52**	**−18.2**
		Landings	1.6	0.93	0.089	0.05	−42.8
Shetland	**49.0**	**Biomass**	**9.4**	**8.8**	**0.46**	**0.43**	**−6.1**
		Landings	0.69	0.51	0.034	0.025	−25.8
Irish Shelf	**121.8**	**Biomass**	**11.8**	**10.1**	**1.4**	**1.2**	**−14.5**
		Landings	1.5	0.93	0.18	0.11	−36.4
Irish Sea	**48.9**	**Biomass**	**12.3**	**10.1**	**0.60**	**0.49**	**−17.9**
		Landings	1.7	1.1	0.084	0.054	−36.3
Celtic Sea	**198.3**	**Biomass**	**11.1**	**10.2**	**2.2**	**2.0**	**−7.8**
		Landings	1.2	0.87	0.24	0.17	−29.2
Armorican	**73.2**	**Biomass**	**12.4**	**9.5**	**0.91**	**0.70**	**−23.1**
		Landings	1.9	1.1	0.14	0.082	−41.5
North Sea	**421.0**	**Biomass**	**12.0**	**9.5**	**5.1**	**4.0**	**−20.1**
		Landings	1.68	1.02	0.71	0.43	−39.2
NW European Shelf	**1039.0**	**Biomass**	**11.8**	**9.8**	**12.2**	**10.2**	**−16.8**
		Landings	1.54	0.97	1.6	1.0	−36.9
Open Ocean	**2377.6**	**Biomass**	**6.0**	**6.8**	**13.1**	**15.0**	**13.8**
		Landings	0.36	0.34	0.79	0.75	−5.0

chlorophyll and oxygen (Fig. S6). The relative proportion of phytoplankton functional groups compared to observational data also appear similar between the two simulations, although the observational data only covers one month for two years (Fig. S7). Validation has not been undertaken for any zooplankton groups due to the lack of available carbon biomass data.

3.2. Impacts of two-way coupling on fish biomass and landings

3.2.1. Fish biomass

Two-way coupling resulted in a decline in fish biomass compared to one-way coupled predictions for all regions of the North West European Shelf (Fig. 3C; Table 2). Across the shelf, the mean fish biomass over 1986–2017 was reduced from 11.8 (1.6 to 29.2 min/max) gWM m^{-2} in the offline HTL simulation to 9.8 (1.3 to 24.3) gWM m^{-2} in the two-way coupled simulation. The decline in fish biomass across the shelf was due to lower fish biomass in the summer in the two-way coupled simulation, while the winter fish biomasses were similar between the two simulations (Fig. 2A). In contrast, in the open ocean fish biomass was greater in the two-way coupled simulation compared to the offline HTL simulation, increasing from 6.0 gWM m^{-2} in the offline HTL simulation to 6.8 gWM m^{-2} in the two-way coupled simulation.

The impact of two-way coupling on fish biomass was not uniform across the North West European Shelf. In the offline HTL simulation mean fish biomass was greatest in the Southern North Sea (13.6 gWM m^{-2}), followed by the Channel (13.5 gWM m^{-2}) and Central North Sea (13.3 gWM m^{-2}) (Table 2). In contrast in the two-way coupled simulation fish biomass was greatest in the Channel, followed by the Central North Sea and Celtic Sea. The greatest impact of two-way coupling was observed in the Southern North Sea, where fish biomass decreased by 32%, whereas in areas such as Shetland and the Celtic Sea fish biomass only decreased by 6 and 8% respectively. On a cell-by-cell basis two-way coupling did not always result in a decline in fish biomass on the shelf, with fish biomass in some cells such as the middle of the North Sea and Celtic Sea increasing by a maximum of 2.2 gWM m^{-2} (in comparison to a maximum decline of 8.4 gWM m^{-2}). Two-way coupling also generally resulted in a smaller proportion of large fish on the shelf and higher proportion in the open ocean (Fig. 3I; S3).

3.2.2. Fish landings

Two-way coupling resulted in lower fish landings across the North West European Shelf than for the offline HTL simulation; fish landings decreased by over one third from 1.5 g WMm^{-2} to 0.97 gWM m^{-2} in the offline HTL and two-way coupled simulations respectively (Table 2). On the shelf, two-way coupling had a relatively larger impact on fish landings than fish biomass with landings decreasing by 37% across the shelf in comparison to a 17% decrease in fish biomass. Conversely in the open ocean, opposite trends occurred between biomass and landings with biomass increasing by 14% and landings decreasing by 5%, reflecting the difference in size structure of fish.

3.3. Impact of two-way coupling on plankton and carbon cycle

Two-way coupling impacted the entire LTL model with different impacts seen between the open ocean and shelf regions (Fig. 5, S8). The impacts were greatest for mesozooplankton biomass, which declined on the shelf and increased in the open ocean in the two-way coupled simulation compared to the baseline LTL simulation (Fig. 5G). The changes in mesozooplankton dynamics resulted in a decrease in carbon cycling within the entire ecosystem (Fig. 6): fluxes into/out of the mesozooplankton pool were largely responsible for the decrease in carbon cycling between the two-way coupled simulation and the baseline LTL. On the shelf, the decline in mesozooplankton biomass, in addition to the removal of intraguild predation and mortality, resulted in less DOC and POC production by mesozooplankton (Figs. 6 and 7) and weakened the microbial loop (Fig. 6). A similar reduction in carbon cycling was simulated in the open ocean despite the increase in mesozooplankton biomass (Figs. 6 and 7).

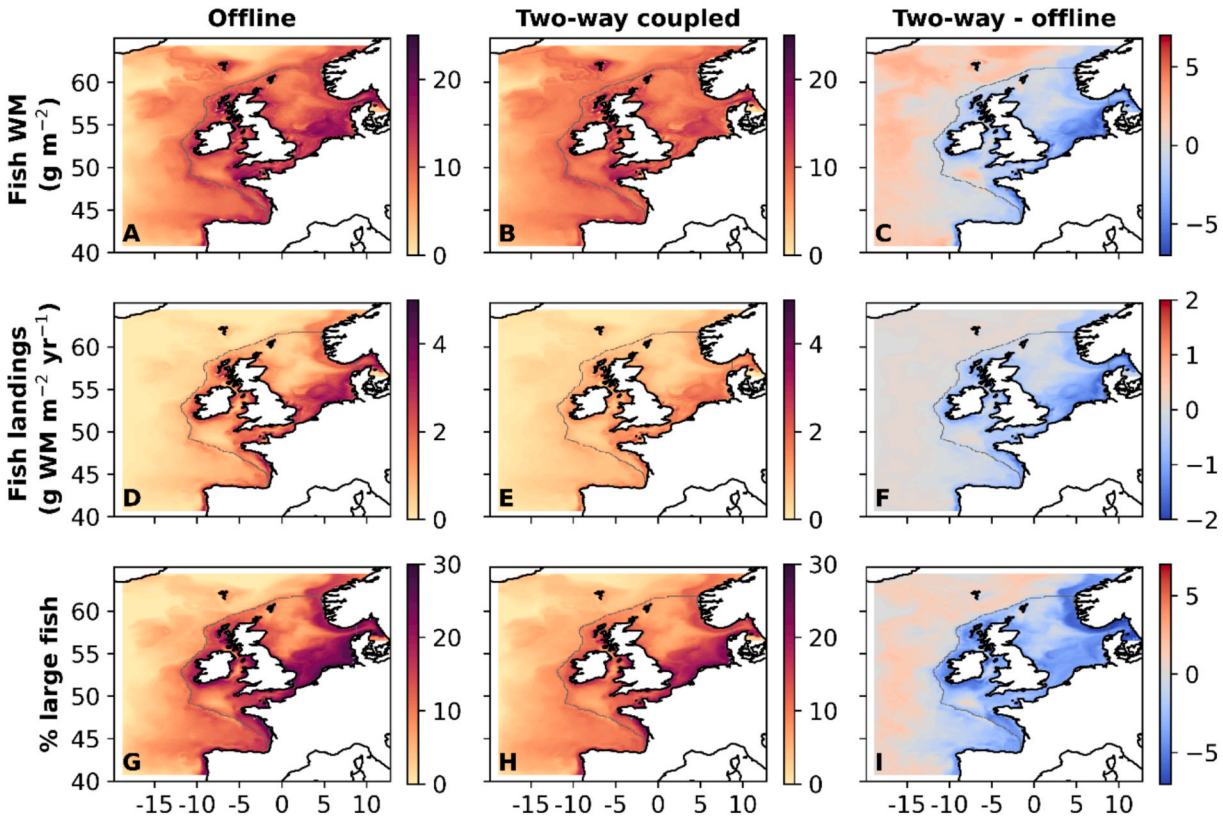


Fig. 3. Comparisons in mean simulated fish wet mass (A–C), landings (D–F) and percentage of total fish classified as large (G–I) over 1986–2017 for the offline HTL simulation (left column), two-way coupled simulation (centre column) and difference between two-way coupled and offline HTL simulation (right column). Large fish were classified as all fish over 80 g. In the difference plots (C,F + I) red/blue colours indicate an increase/decrease in the two-way coupled simulation compared to the offline HTL simulation. The grey line indicates edge of the North West European Shelf.

Such changes in the pelagic carbon cycle consequently affect the potential sequestration of carbon by the marine system: the net seafloor POC flux decreased on average in large parts of the shelf, where the magnitude of the POC flux is higher (Fig. 5I), which impacts carbon processing and storage in the sediments. In addition, the North West European Shelf becomes less efficient at absorbing atmospheric CO_2 with the air-sea CO_2 flux declining by 7% from 29.3 to 27.2 $\times$ 10^6 tC yr^{-1} in the baseline LTL and two-way coupled simulations respectively (Fig. 5H).

The large changes in mesozooplankton biomass upon two-way coupling was predominantly a result of changes in predation pressures on the mesozooplankton and removal of natural mortality. The replacement of mesozooplankton intra-guild predation with predatory pressure from fish results in spatial differences in the predatory pressure of the mesozooplankton (Fig. 8). The predatory pressure from fish on mesozooplankton was generally higher than intra-guild predation throughout the domain with the exception of the Southern North Sea (Fig. 8A, B and C). However, due to the removal of natural mortality, the overall mortality rate (natural + predatory mortality) for mesozooplankton in the two-way coupled simulation was lower than for the baseline LTL simulation across the entire domain on annual timescales (Fig. 8). Despite this, opposing trends in annual mesozooplankton biomass occurred between the shelf and open ocean in the two-way coupled simulation in comparison to the baseline LTL simulation.

The change in predation pressure of mesozooplankton also impacted their phenology. Despite a 17% decline in mesozooplankton biomass across the shelf, there was an increase in the maximum annual peak of mesozooplankton biomass across large sections of the shelf (Fig. 9). Strong increases (>45%) were observed in the middle of the North Sea and Celtic Sea, while decreases in the maximum peak occurred in areas close to the coast, the German Bight and English Channel. In addition, the mesozooplankton peak biomass occurred up to a month earlier throughout the shelf in the two-way coupled simulation (Fig. 9).

All plankton groups were impacted by two-way coupling, although to a lesser extent than mesozooplankton (Fig. 5). Opposing trends were predicted between different size classes: the large phytoplankton (diatoms, micro) biomass decreased as a consequence of two-way coupling whilst the smaller phytoplankton (pico and nano), which were too small for fish to predate on, typically had a higher biomass in the two-way coupled simulation as a result of a lower grazing pressure from the zooplankton. Trends in microzooplankton were less clear due to both being predated on by fish and having reduced grazing pressure from mesozooplankton (Fig. 5F and G). However, in general the microzooplankton followed similar trends to the equivalent size of phytoplankton.

4. Discussion

There is a growing interest in the impact of fish and the role of anthropogenic fishing activities on the global carbon cycle (Andersen et al., 2024; Cavan & Hill, 2022; Martin et al., 2021, 2023). However, in order to accurately quantify these impacts, spatially varying hydrodynamic-ecosystem models with two-way coupling to fish need to be used. Note that the hydrodynamics can be provided offline, if a high enough resolution is saved to adequately resolve the biogeochemistry. The model framework presented here provides a coherent and fully coupled system that allows fish to be modelled in a more realistic manner that more rigorously represents both bottom-up and top-down impacts of environmental change on marine ecosystems. The simulations demonstrate that switching from one-way to two-way coupling between biogeochemical and fish models changes estimates of the ocean's ability to absorb CO_2 and estimates for fish biomass and

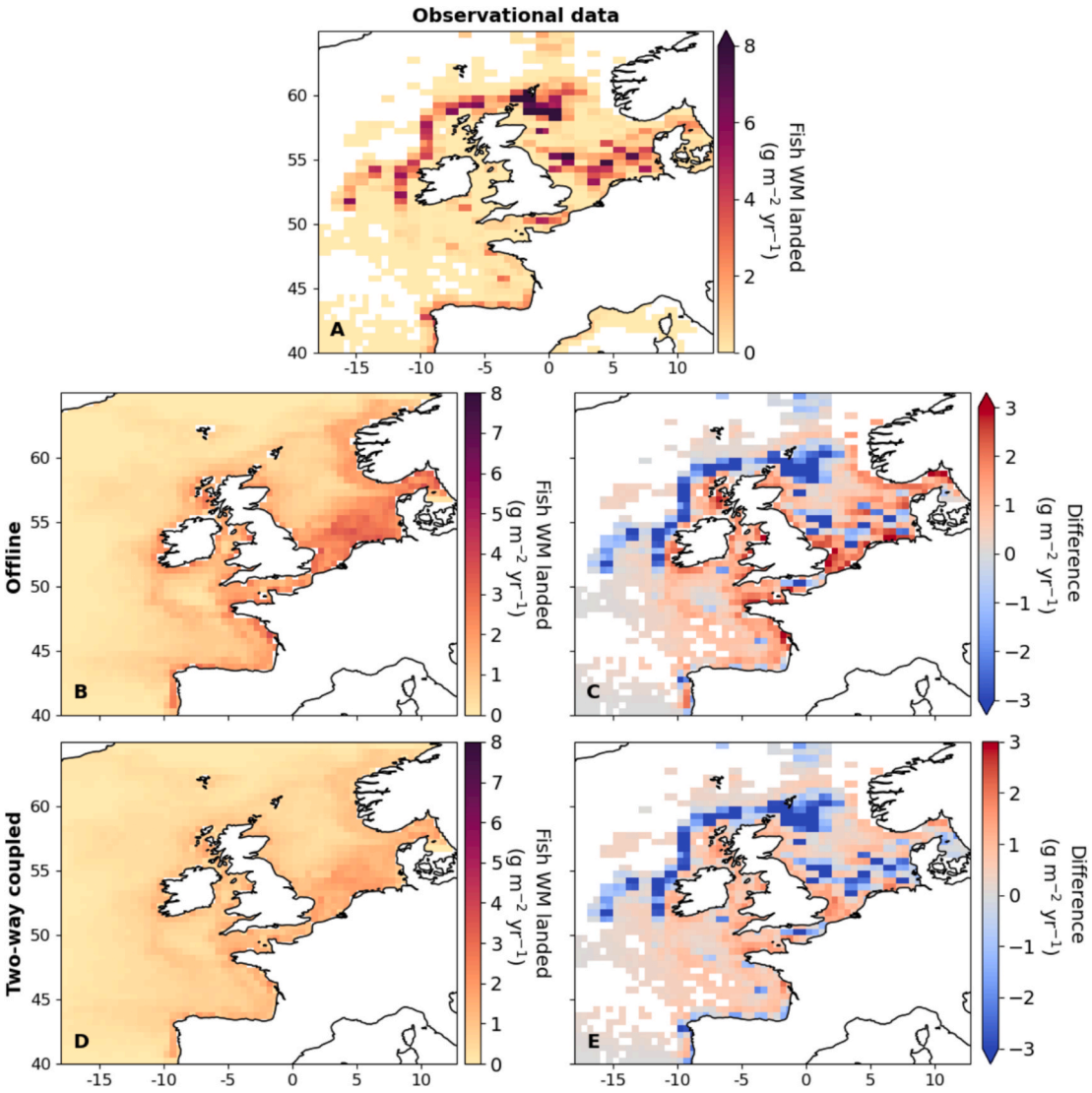


Fig. 4. Comparison of (A) observed pelagic fish landing data (Kovšars et al., 2022) over 2014–2017, fish landings in the (B) offline and (D) two-way coupled simulation over same time period and difference between the (C) offline/E) two-way coupled simulations and observational data. Note that the model simulations have been regridded onto the same grid as observed landing data.

landings with implications for predictions on the impacts of climate change and fisheries management on fish communities.

A unique feature of this modelling framework is the ability to produce more realistic high resolution depth-resolved feedbacks between LTLs and fish, in addition to being able to produce high spatial resolution predictions of fish biomass. While validation does not necessarily show an improvement in model predictions the aim of this work is to show the impact of top-down and bottom-up feedbacks propagating through the system which are missed in both LTL or HTL modelling studies. This big benefit comes with a limited increase of the computational cost, with the two-way coupled simulations taking only about 10% longer to run than the baseline LTL simulation. Consequently, it is not detrimental to start including fish in standard model runs.

4.1. Fish impacts on the carbon cycle

Two-way coupling predominantly impacts mesozooplankton biomass within the LTL model. Top-down effects result in impacts cascading down throughout the planktonic food web influencing the size structure of phytoplankton, in addition to predictions for carbon sequestration such as the quantity of carbon reaching the sea floor and predictions for air-sea fluxes of CO_2 (Fig. 5).The predicted increase in small phytoplankton and decrease in larger phytoplankton shows the sensitivity of these groups to top-down impacts. This is important as climate change predictions indicate that phytoplankton will shift towards smaller cell sizes in the future (Flombaum et al., 2013; Henson et al., 2021). However, these climate change models typically do not include any interactions with fish. Changes in fish stocks as a consequence of anthropogenic fishing pressures or increasing temperatures may alter top-down cascades through the planktonic food web and thus future predictions of size structure of the food web (Blanchard et al., 2012; Petrik et al., 2020). Therefore, getting the right predation pressures on plankton from fish is important in order to predict current and future trends on size structure of the marine ecosystem, in addition to improving our understanding of the global carbon cycle.

The switch to two-way coupling resulted in a substantial decline in the predicted air-sea CO_2 flux and in carbon reaching the seafloor. The change is predominantly a result of changes in mesozooplankton biomass producing less POC and DOC and reducing strength of microbial loop. The decline in the air-sea CO_2 flux across the North West European Shelf of 2.0×10^6 tC yr^{-1} is greater than the change predicted due to the inclusion of terrestrial DOC in model simulations (1.3×10^6 tC yr^{-1};

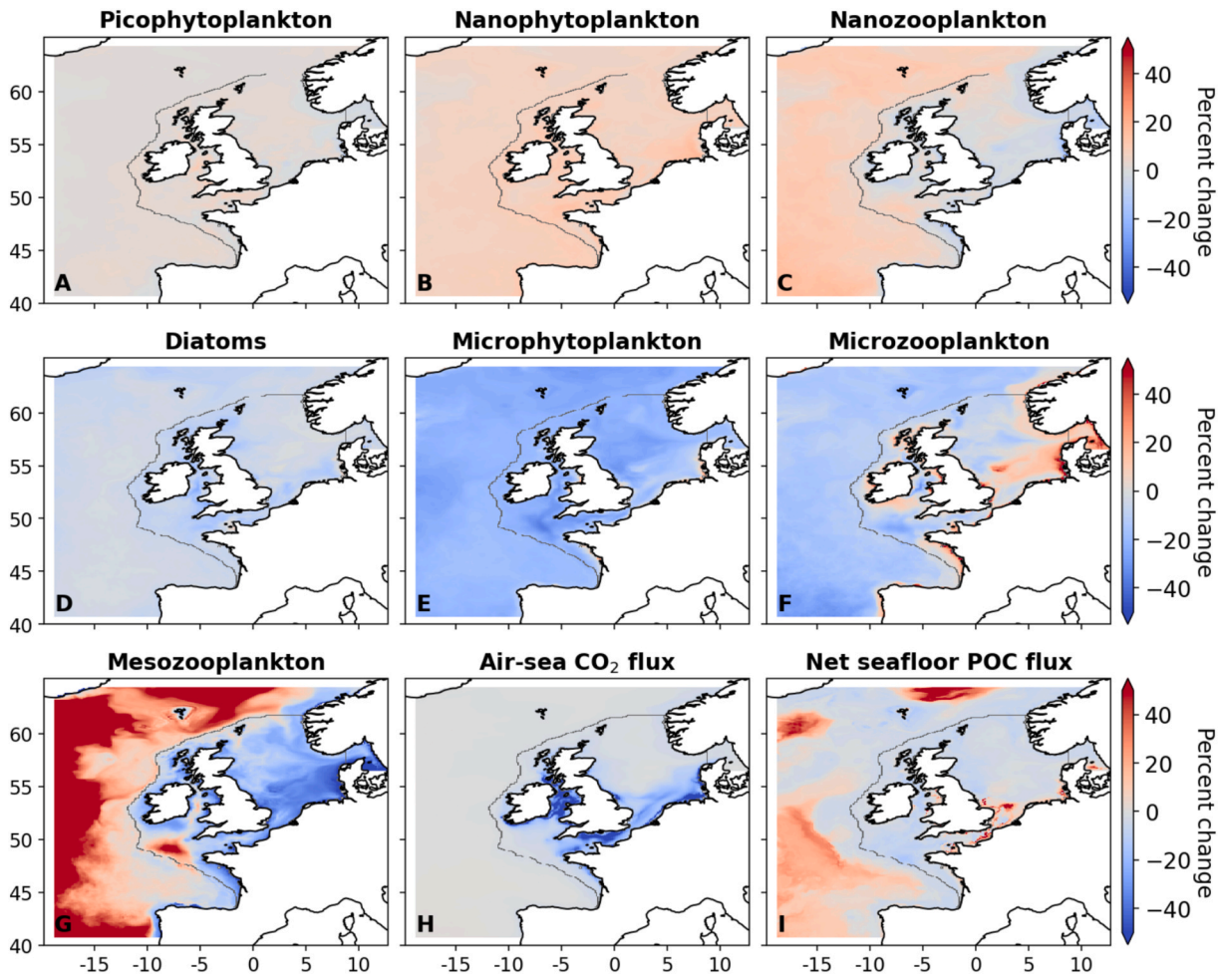


Fig. 5. Impact of two-way coupling on ecosystem properties compared with one-way coupling (baseline LTL simulation) on simulated size classes of depth-integrated phytoplankton and zooplankton (A–G), air-sea CO_2 flux (H), and net seafloor POC flux (I) over 1986–2017. A positive change (red) indicates that two-way coupling increases the value of the variable shown. The grey line depicts the shelf edge.

Powley et al., 2024). It is also much greater than the fish landings removed from the shelf of 0.97×10^6 tC yr^{-1} (Table 2), highlighting the sensitivity of this flux to changes in ecosystem structure. However, despite the change, the total predicted air-sea flux of 29.3×10^6 tC yr^{-1} is still well within the uncertainty of predictions of $15.6–39.6 \times 10^6$ tC annually (Kitidis et al., 2019; Legge et al., 2020). In addition to changes in the air-sea fluxes, adding the feedbacks of fish predation and excretion on the pelagic system resulted in a reduction of the POC flux to the seafloor of about 2%. While this may seem small there is substantial variability in the degree of magnitude, and even direction of this flux, meaning regionally it can be important. This has consequences on the benthic community and its ability to process organic carbon, with wider consequences on carbon sequestration and oxygen consumption in the sediments (Fig. S8). Impacts on the benthic community are expected to be even greater if fish feeding on the benthos was represented in the model.

4.2. *Mesozooplankton mortality: a key process with little constraints*

Zooplankton mortality is a key linkage between higher and lower trophic level models. It has long been recognised as a sensitive and uncertain process within lower trophic models with impacts on the entire ecosystem (e.g. Edwards and Yool, 2000; Mitra et al., 2014; Rohr et al., 2023; Steele and Henderson, 1992). This has also been observed within two-way coupled models with changes in the zooplankton predatory mortality controlling direct and indirect top-down effects throughout the plankton spectrum (Travers et al. 2009).

A key benefit of our modelling approach is a more realistic representation of zooplankton mortality in ERSEM, which in its standard form (Butenschön et al. 2016) does not fully reflect the predation on zooplankton due to fish population. The issues with the standard mortality formulation were demonstrated through two-way coupling between MIZER and ERSEM requiring substantial changes to select model parameters in order for them to produce reasonable results. Volumetric clearance rates were reduced for fish compared to those in Blanchard et al., (2009), while mortality terms and intra-guild predation are turned off the mesozooplankton, to ensure the survival of mesozooplankton and consequently fish. The lack of feedbacks in offline/one-way coupled simulations allows HTL models to use a higher clearance rate and consequently produces higher predatory and growth rates for fish than that what would otherwise occur: fish can continually predate on plankton with no impact on their prey enhancing their growth rates and biomass. The required changes to parameters thus suggest that descriptions of mortality in the stand-alone LTL model are incompatible and inconsistent with predation pressures predicted by the HTL model.

The requirement to remove natural mortality for mesozooplankton in ERSEM upon two-way coupling is unrealistic as these processes are thought to be unrelated to predators such as fish: they are likely to occur in nature due to disease and senescence. Such an unrealistic change was not needed in other two-way coupled models where this natural mortality term for mesozooplankton is still present, but it is significantly reduced compared to the one-way coupled implementation. Reductions

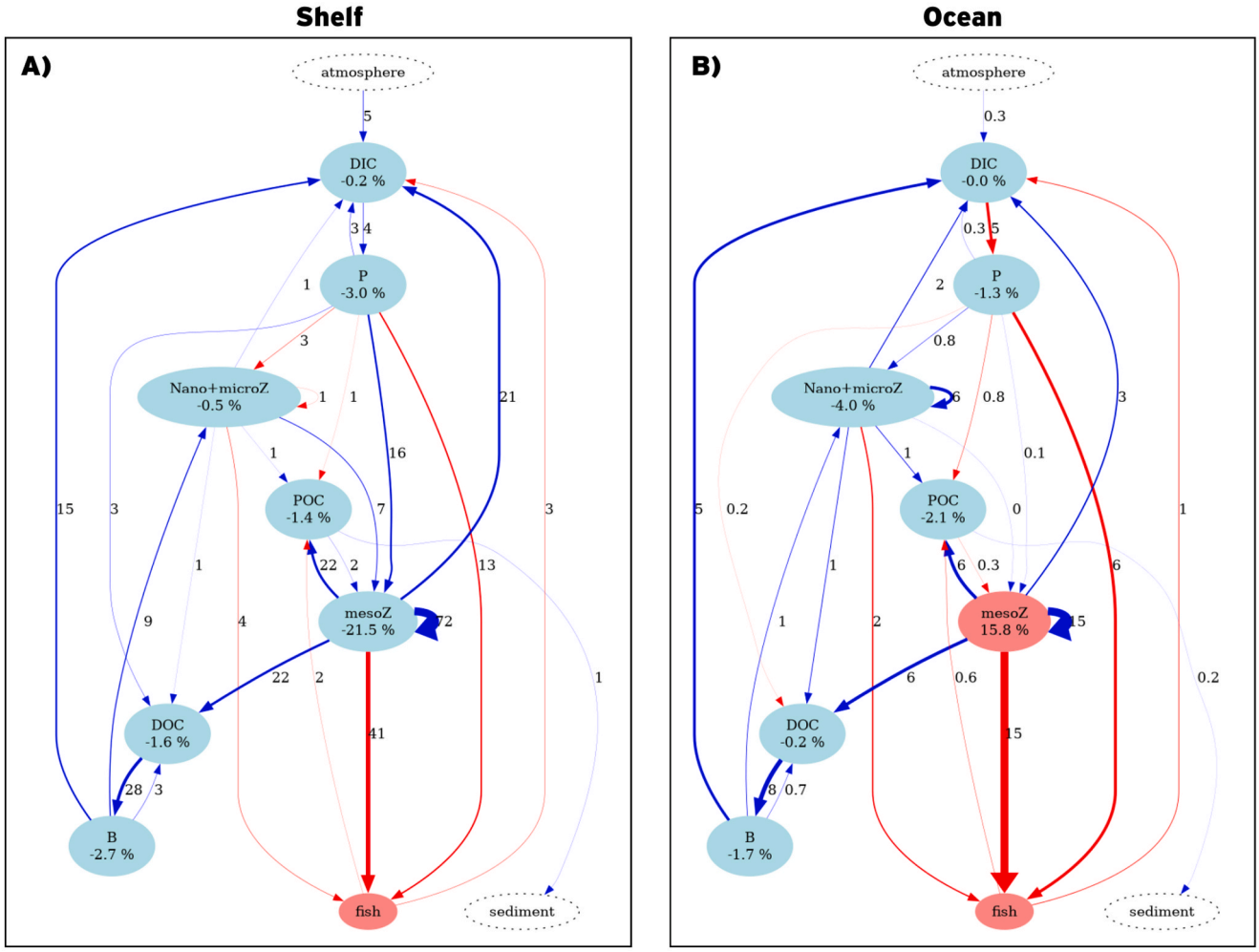


Fig. 6. Flux diagram showing the difference in mean fluxes (mg C m^{-2} d^{-1}) and carbon pools (%) between the two-way coupled simulation and baseline LTL simulation over 1986–2017 for (A) the shelf and (B) the open ocean. The weight of the arrow is proportional to size of the difference. Blue arrows show a decrease of the flux in the two-way coupled model compared to the baseline LTL model, while red arrows highlight an increase. Fluxes for fish are all positive as this shows additional impact fish has on the LTL model and fluxes for offline HTL simulation are not included. Note POC uptake by bacteria is missing from this figure due to the flux not being saved in the simulation.

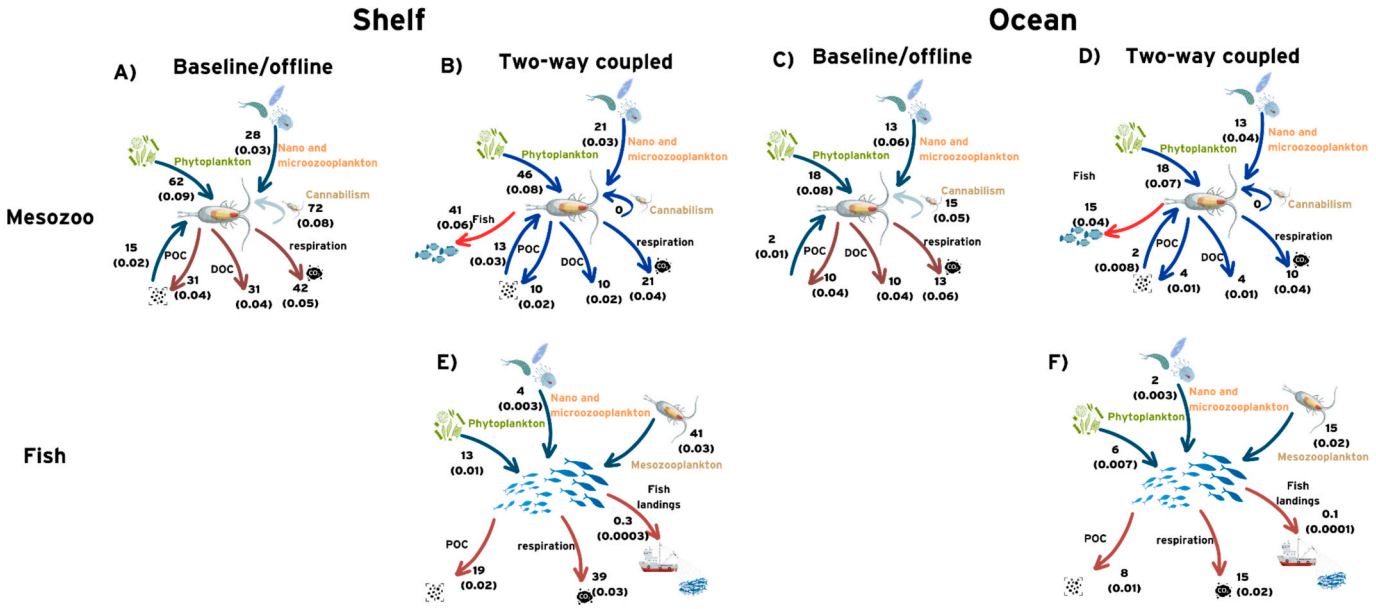


Fig. 7. Simulated mass balances for mean mesozooplankton and fish biomass on the North West European Shelf (A, B and E) and open ocean (C, D and F) in the baseline LTL and two-way coupled simulations over 1986–2017. Numbers represent mean depth integrated flux (mg C m^{-2} d^{-1}), numbers inside brackets represent the specific rate (flux divided by mesozooplankton biomass (A–D) or fish biomass (E + F), in units d^{-1}). Numbers are missing for the offline HTL simulation as fluxes were not saved.

vary from an order of magnitude lower (NEMO-PISCES-APECOSM, Aumont et al., 2018), to 20% (ECOSMO-EWE; Daewel et al., 2019) while in OSMOSE natural mortality is based on a function of the availability constant of mesozooplankton to fish (Travers et al., 2009; Banaru et al., 2019). Natural mortality contributes to a median of 4.4% of the total annual mortality of the copepod *Calanus helgolandicus* at a station in the

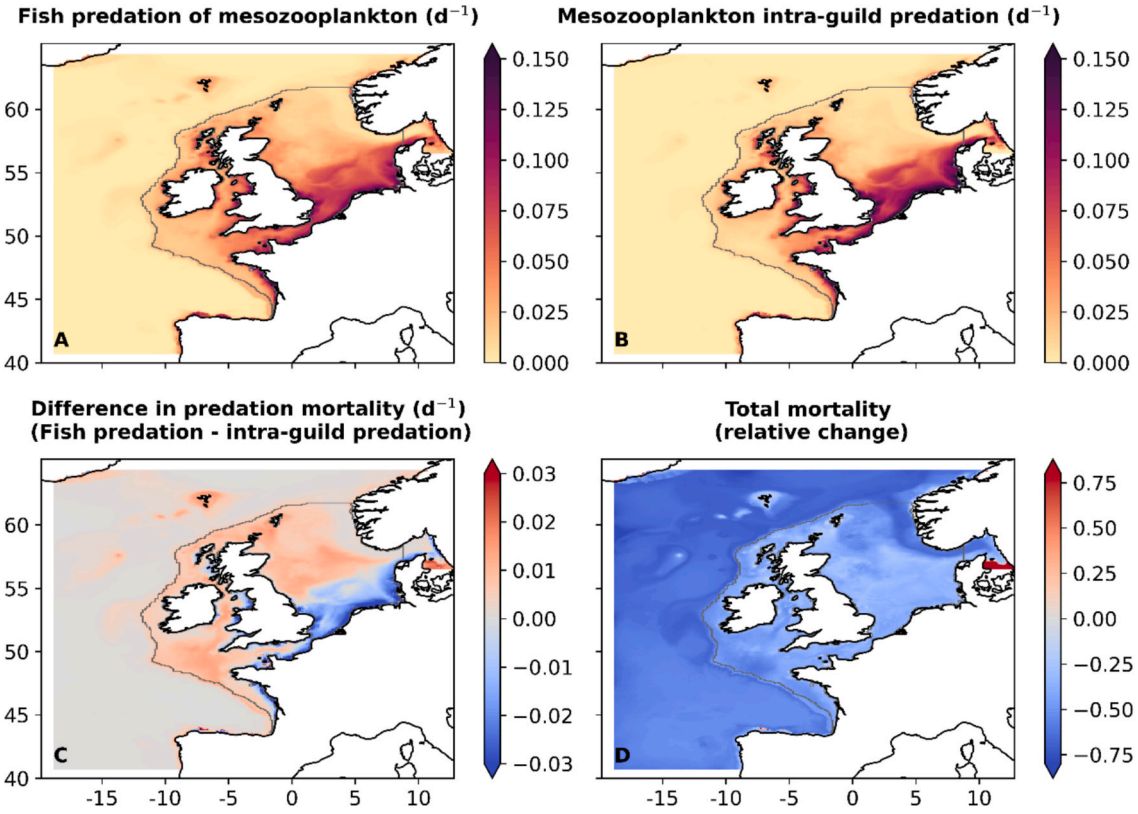


Fig. 8. Changes in mean water column predation rates on mesozooplankton due to two-way coupling. (A) fish predation on mesozooplankton (two-way coupled simulation), (B) mesozooplankton intra-guild predation (baseline LTL simulation, zero in two-way coupled simulation) (C) difference between fish predation of mesozooplankton in the two-way coupled simulation and mesozooplankton in the baseline LTL simulation and (D) the mean total mortality (natural + predation) difference between two-way coupled and baseline LTL simulation over the course of 1986–2017. The blue/red colour indicates a decrease/increase in mortality in the two-way coupled simulation in comparison to the baseline LTL simulation.

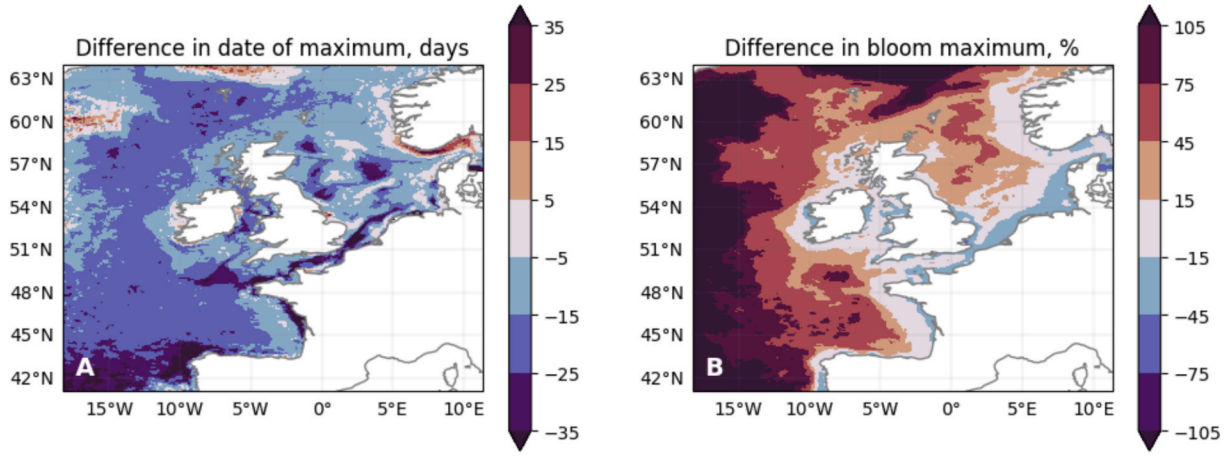


Fig. 9. Differences between simulated mesozooplankton phenology between the two-way coupled model and baseline LTL simulation. (A) Difference in the date of mesozooplankton maximum biomass, and (B) percentage difference in the depth integrated maximum mesozooplankton biomass between the two-way coupled and baseline LTL simulation. The blue colours indicate an earlier mesozooplankton peak and smaller peak in the two-way coupled simulation in comparison to the baseline LTL simulation.

English Channel (Maud et al., 2018) while it is estimated to contribute to 25%–33% of total mortality of copepods on a global scale (Hirst and Kiorbe et al., 2002). The large variability in mortality in both experimental and model data highlights how a stronger experimental constraint on natural mortality is much needed.

Mesozooplankton predation mortality is greater in this study than other estimates which contributes to the need to remove natural mortality. For example, mean predation mortality across the North Sea of 0.048d^{-1} (range: 0.008–0.19 d^{-1}) predicted in this study is greater than that predicted using ECOSMO-EWE of 0.03 d^{-1} (Daewel et al. 2019) and from a combination of observational and model results (~0–0.04 d^{-1} seasonally; Maar et al. 2014). In total our estimate for average predation mortality of mesozooplankton of 15.5 gC m^{-2} yr^{-1} for the North Sea is almost 3 times greater than that predicted by Maar et al. (2014) (5.7 gC m^{-2} yr^{-1}) and our estimate also exhibits a much greater spatial variability with a standard deviation of 9 gC m^{-2} yr^{-1} compared to 0.7 gC m^{-2} yr^{-1}. One reason for the different estimates in mortality rate between the studies is that predation mortality is stopped during winter when mesozooplankton biomass is too small in Maar et al., (2014). Similarly, other models limit the amount of zooplankton that the fish can predate on either via an availability constant of zooplankton to fish (OSMOSE) or preference function (ECOSMOII-EWE). In the model setup used here, the predation mortality of plankton is solely dependent on size and occurs year-round. The current parameterisation of the fish ingestion rate (Equation S6) allows for unlimited feeding on prey without satiation. The shape and parameterisation of the feeding functional response has been shown to be important for fish (Flynn et al., 2021; Fulton et al., 2003). A capped response may increase mesozooplankton concentrations and could allow the addition of natural mortality and/or some intra-guild predation of mesozooplankton into model formulations. Therefore, further consideration of feeding parameters should be considered in future model iterations.

Finally, the temporal variability in mesozooplankton mortality is higher in the two-way coupled simulation driven by a bigger seasonality of fish biomass than that predicted by other modelling studies (Daewel et al., 2019). The increased seasonality of fish biomass is likely due to the lack of predation on benthic food sources by fish. Planktonic prey has indeed a marked seasonality that is much less evident in benthic fauna (Lessin et al., 2019) which can be the largest component of fish diet in winter months (up to 85%, Daewel et al., 2019) when pelagic food sources are at their minimum. The non-modelled demersal fish will therefore have less seasonal variability than pelagic fish (Greenstreet et al., 1997). Thus, the inclusion of benthic food sources would allow a more stable seasonal pattern of fish biomass and therefore a reduced variability in mesozooplankton mortality.

The change in predatory pressure on mesozooplankton with two-way coupling results in temporal changes in mesozooplankton biomass between the two simulations. In the baseline LTL simulation, intraguild predation is proportional to mesozooplankton biomass so a temporal offset between growth and predatory pressures is not present: intra-guild predation continuously suppresses the mesozooplankton population growth rate, and thus immediately dampens the mesozooplankton response to an increase in prey availability. In comparison in the two-way coupled simulation, the size classes of fish feeding on mesozooplankton take time to develop from the smallest ("fish egg") size class that closely tracks mesozooplankton. Accordingly, a lag between mesozooplankton growth and fish predation occurs, alleviating predatory pressure on the mesozooplankton and consequently resulting in a more rapid increase in mesozooplankton biomass early in the season and greater peak biomass across the majority of the shelf (Fig. 9). However, by the time the size classes of fish that feed on mesozooplankton catch up, the predation pressure they exert exceeds that of the original intraguild predation formulation. Accordingly, the earlier and more pronounced mesozooplankton bloom is then quickly grazed down by the fish, leading to the decline in mesozooplankton biomass.

The sensitivity of mesozooplankton to two-way coupling and the parameterisation of fish search rates within the HTL highlights the need to rigorously validate the mesozooplankton group within the LTL model. We show that the model is sensitive to clearance rate parameters and hence larger changes may be observed if different values for these parameters were used. Spatial and temporal data from the continuous plankton recorder (CPR) could be used to validate the model, especially within the North Sea (Holland et al., 2024). However, in order for CPR data to be relevant to models such as ERSEM, rigourous methods need to be employed to convert abundance to biomass in units of carbon, which has significant uncertainty associated with it. In addition, CPR data only captures data on the near-surface so other validation methods are needed to validate the mesozooplankton biomass throughout the water column.

This work also highlights the need to rethink how top closure terms are implemented in LTL models without HTL model coupling. A previous study suggested that intraguild predation of mesozooplankton should be used over traditional quadratic, hyperbolic or sigmoidal closure terms (Mitra, 2009). However, our results suggest that a density dependent top closure term may be more representative of fish predation and HTLs than intra-guild predation which consequently impacts the rest of the carbon cycle. Alternatively, Maar et al. (2014) implemented a variable mesozooplankton mortality within an LTL model

based on observational maps of zooplanktivourous fish, model of fish larvae and observed ingestion rates. The use of observational data in Maar et al.,'s method, however, makes their work unsuitable for different regions and future projections. The model presented here could be used to infer relationships between mesozooplankton and fish biomasses that predate on the mesozooplankton to parameterise the top closure term. This could vary from calibrating more simplistic quadratic mortality term to creating both a spatially and temporally changing top closure term, with the caveat that the more complex/detailed the term becomes the more regionally specific the closure term becomes.

4.3. Heterogeneity in spatial impacts of two-way coupling

Two-way coupling does not produce spatially uniform changes in either plankton or fish biomass and landings across the domain. Clear differences were observed in both the response of fish and mesozooplankton biomass between the shelf and the open ocean with both decreasing on the shelf and increasing in the open ocean. However, the reasons for this are unclear. The replacement of intra-guild predation as a top closure term for mesozooplankton with fish predation makes disentangling the reasons for the opposing trends in mesozooplankton biomass between the shelf and the ocean difficult. Intra-guild predation not only affects mesozooplankton mortality, it also acts as a food source to the mesozooplankton contributing to 44% of total ingested prey on the shelf in the baseline LTL simulation versus 32% in the open ocean (Fig. 7). Removal of this food source removes a substantial amount of available food to mesozooplankton, especially on the shelf. The higher ingestion due to intra-guild predation also accounts for additional losses through respiration, excretion and egestion on the shelf and in the ocean than if this was not present, which is key to driving the internal carbon cycle of the system on both the shelf and in the open ocean. As mesozooplankton represent the largest planktonic group available to fish, the changes in fish biomass mirrored the changes in mesozooplankton biomass.

It was not only shelf vs open ocean which showed non-uniform changes, however. Different areas of the shelf responded differently to two-way coupling for both fish and plankton groups (Figs. 3 and 5, Table 2). The productive Southern North Sea was particularly sensitive to these feedbacks, emphasising increased model uncertainty in this region when two-way coupling is not included. In addition, relative changes between the two-way coupled and offline HTL simulation are not the same for fish biomass and fish landings highlighting changes in the size structure of fish. On the shelf, fish landings were more sensitive to two-way coupling than fish biomass declining on average by 37% versus 17% for fish biomass. Similarly, fish landings decreased in the open ocean despite increases in fish biomass. The reduction in the proportion of (and total) large fish (Fig. 3, S3 and S4), here defined as fish weighing over 80 g WM, on the shelf leads to the lower fish catch. The lower proportion of large fish also suggests that there is reduced trophic transfer throughout the fish population and a steeper size spectrum within the ecosystem. A reduction in large fish is also predicted in global climate projections due to a reduction in zooplankton populations and increased metabolic costs (Petrik et al., 2020). Our result suggests that two-way coupling may further amplify these predictions in the prediction of large fish. Overall, the non-linear feedbacks across the shelf to two-way coupling suggest it is not possible to parameterise the impact of two-way coupling in models using a simple linear equations and top closure terms. Fully prognostic models are needed in order to accurately predict spatially resolved fish biomasses and landings.

4.4. Model caveats

This is the first study to two-way couple ERSEM, a HTL model representing a community of fish and a state-of-the-art 3D hydrodynamic model. However, improvements can be made to the current model set-up. Currently the two-way coupled model only includes feeding on pelagic prey, with fish having no interaction with the benthic system. Therefore, modelled fish biomass excludes demersal fish. Demersal fish comprise approximately 20–25% of fish catch on the shelf (Kovšars et al., 2022) so this lack of coupling may reduce the predictive capabilities of the model. Macro-benthos represents between 32–38% of community (pelagic + demersal) fish diets in the North Sea (Greenstreet et al., 1997). Consequently, an important dietary component is missing from fish diets in this study.

Future evolutions of the model should consider including spatial migration of the fish, with fish staying within their 7 km × 7 km grid area in this study. Consequently, fish biomass in a given cell is entirely dependent on the resource spectrum of the pelagic plankton within that cell. While in coarser grids or boxes this assumption is reasonable it becomes more questionable the higher the resolution of the model domain. This problem is not unique to FABM-ERSEM-MIZER with other two-way coupled models such as ECOSMOII-EWE (Daewel et al., 2019) also not including migration of fish. Multiple methods to represent horizontal movement exist, such as restricted-area search, random walk, and run and tumble (Watkins and Rose, 2013 and references therein), but careful consideration of the method and substantial testing will be required. In addition, the current FABM framework needs to be expanded to support advection/diffusion-style parametrisations of horizontal movement, along with the capability to obtain horizontal gradients in environmental variables and prey needed for such parameterisations. The lack of migration of fish in addition to the constant fishing pressure over space and time partially explains the discrepancies between observed landing data and model predictions. Fish caught along the North Western Shelf edge are predominantly migratory fish such as mackerel and blue whiting (Kovšars et al., 2022). Inclusion of fishing pressure maps may partially mitigate some of these issues.

A critical challenge in fisheries modelling is the use of different units across modelling studies and in observational studies. Here, we used a constant carbon:wet mass ratio of 0.1 (gC:gWM) in FABM-MIZER for all fish. Conversion factors produce inherent uncertainty when comparing model results with both data and other models with due to variations in the conversion factor by up to a factor of four (0.045–0.184 gC:gWM; Daewel et al., 2019; Heath, 2012; Petrik et al., 2019) and some models varying conversion factor on species type whilst others use a constant value. Considering fish catches are reported in g wet weight, a community wide consensus needs to be made on conversion approaches in order to be able to accurately and reliably compare model results and data which will also help decision makers understand uncertainty in model results.

The model used here uses a simple stoichiometric Redfield ratio for fish. However the internal nutrient stoichiometry for fish is also variable, with fish often rich in phosphorus due to the presence of phospholipids. There is increased recognition of the impact of fishing on the biogeochemical cycles of phosphorus and nitrogen in the ocean (González Ortiz et al., 2025; Le Mézo et al., 2022; Ray et al., 2025). This study focuses on the fate of carbon, but future studies could investigate impacts to the nitrogen and phosphorus cycles. Analysis using the Redfield ratio would give a lower bound to the amount of phosphorus removed due to fishing, but in order to accurately investigate these impacts, improved nutrient ratios within fish will be needed.

Climatic and anthropogenic pressures may cause the collapse of individual fish stocks. However, FABM-MIZER is run using the community model so inferences about particular groups of fish cannot be made nor can the collapse of fish stocks of a particular species. This is because in a community-level model the fish stock may be assumed to be replaced by another species of fish and thus total biomass may not change. MIZER can be run using multispecies mode (Scott et al., 2014; Blanchard et al., 2014) but to simulate multiple species using FABM-MIZER a predation matrix will need to be implemented. Nonetheless, the model can be used to understand large scale changes in fish dynamics in addition to improving the closure term in LTL models.

5. Conclusions

Our work highlights that mesozooplankton are the most sensitive group affected by the implementation of two-way coupling between a LTL marine model and a HTL model. Both average biomass and seasonal dynamics are significantly changed by adding the top-down control from fish. A substantial reparameterization of both models is required, in particular of the clearance rates of the HTL model and of the natural mortality of mesozooplankton in the LTL model. Both mesozooplankton mortality and fish feeding need to be carefully and simultaneously calibrated in a two-way coupled simulation to ensure coexistence of zooplankton and fish, while retaining ecologically plausible parameterisations. More empirical constraints on these parameters are needed and so is a comparison and intercalibration of these processes among existing models.

Two-way coupling reduces carbon cycling on both the shelf and open ocean in addition to changing food web dynamics and the size structure of plankton groups. The decline in carbon cycling is largely a consequence of changes in mesozooplankton dynamics. The changes result in a reduction in the ocean's ability to absorb carbon from the atmosphere with potential impacts on the global carbon cycle. Thus two-way coupling should be considered when running future climate simulations. While more work is needed to improve the representation of fish dynamics at higher resolution (e.g. movement, benthic feeding), given the limited increase in computational costs (~10%) and the importance of the impacts highlighted by this study, we strongly recommend to include two-way coupling in all future implementations of end-to-end marine modelling systems.

CRediT authorship contribution statement

Helen R. Powley: Conceptualization, Formal analysis, Software, Validation, Visualization, Writing – original draft. **Rebecca Millington:** Writing – review & editing, Visualization. **Yuri Artioli:** Writing – review & editing, Funding acquisition. **Robert J. Wilson:** Writing – review & editing, Validation. **Jorn Bruggeman:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pocean.2026.103701.

Data availability

The model data presented in this manuscript, along with all biogeochemical/fish parameter values can be found at https://zenodo.org/records/15831075. Full model output data is available on request from the authors: hpo@pml.ac.uk.

The code used to run this model is publicly available. The NEMO code can be found at: https://github.com/pmlmodelling/NEMO3.6-FABM, commit PreFABM-498-g2f57d08,

FABM: https://github.com/fabm-model/fabm, commit v1.0.3-35-g98b6334, ERSEM: https://github.com/pmlmodelling/ersem, commit 20.10-9-gf3a7187, FABM-MIZER: https://github.com/pmlmodelling/fabm-mizer, commit 586a0564b2ebd9c6f9ef7c12bdaf7589e84cf528.

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